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GEOODE-KD: DISTILLING SENTENCE EMBEDDINGS AS TEACHER-GUIDED GEOMETRIC DYNAMICS

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ABSTRACT

Table 1: Main results for distillation from Qwen3-Embedding-4B to BERT-base. Classification datasets are evaluated with F1, pair-classification datasets with average precision (AP), and semantic textual similarity datasets with Spearman correlation. Higher is better. Among non-teacher models, the best result is shown in bold and the second-best result is underlined.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Avg.
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>										
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	54.36
Stella	91.08	73.40	64.40	<u>86.23</u>	83.22	<u>68.98</u>	78.38	71.31	79.16	77.35
CDM	89.71	70.76	52.67	84.03	71.47	69.83	71.36	68.87	74.68	72.60
DSKD	88.89	69.31	53.39	86.35	67.73	68.89	70.58	67.99	74.29	71.94
EMO	85.92	69.39	55.41	85.79	64.34	67.73	63.79	64.41	70.14	69.60
TALAS	93.34	74.45	72.70	86.16	87.78	68.11	<u>80.70</u>	77.34	83.57	80.46
GeoODE-KD	<u>92.92</u>	<u>74.05</u>	<u>72.49</u>	85.89	<u>86.12</u>	66.60	80.78	<u>76.01</u>	<u>81.23</u>	<u>79.57</u>

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